Shared failure and the price of verification in multi-LLM systems

Haochuan Kevin Wang
Massachusetts Institute of Technology

Ruijie Shi
Massachusetts Institute of Technology

Abstract

A system built from language-model calls can spend a fixed budget in three ways: more calls whose answers are combined (*width*), more reasoning or a stronger model per call (*depth*), or a check that picks among answers (*verification*). We fit a factor model of item difficulty, system ability and a failure slope to 464 public systems on 1,461 items (SWE-bench Verified, MathArena, LiveBench) and read it beside small preregistered API studies on sealed items. Failures are mostly shared: item difficulty accounts for 0.73 (item-bootstrap interval 0.70–0.75) of the latent variation in success on SWE-bench Verified and 0.73 on MathArena. Given the same calibration data as a one-factor copula, the model predicts how often all 87 held-out SWE-bench systems fail a held-out task (5.9% predicted, 7.6% observed), where the copula underprices it (3.5%); on small math sets both underprice. Across 19 public pairs of one model at two adjacent reasoning settings, one step up raises ability by 0.92 logits and leaves the failure slope unchanged (0.03, -0.14–0.21): the higher setting still fails the items that are hard for everyone. On fresh sealed items, raising a cheap model's reasoning one step beat repeating its no-reasoning call at the nearest measured dollars (confirmatory; 3.7 and 10.7 points, the deeper call costing 1.17× and 1.38× as much); on public pairs such a step pays where a model starts low and about nothing at the top of its range, so depth against width beyond the first step is open. Label-free selection among systems, given the same number of calls, did not beat the best system's own vote. Public-data results are post hoc, and every number carries its status.

1 Introduction

Suppose an agent system has a fixed budget per task. It can ask a cheap model several times and vote, ask a stronger or longer-reasoning model once, or ask several models and let a judge choose. Which should it buy first? The answer depends on one quantity: how much the systems' failures are *shared*. If every system fails the same hard items, extra calls fail with them, and only a system that moves the hard items helps.

Three claims, tested rather than assumed. (i) Failures are dominated by shared item difficulty, so width is capped by the items everyone fails. (ii) Reasoning moves a system along that same difficulty axis, so depth may be the cheaper first purchase. (iii) Picking among agents needs a signal stronger than agreement. On our data, (i) holds on code and math, with a factor model that prices the shared tail on code; (ii) holds as a description of what reasoning does, and as a purchase only for the first effort step of cheap models, which a preregistered test supports; beyond it the question is open and is the subject of a separate preregistered study (Section 4); (iii) holds for agreement-based selection, while our one judge test was capped by its candidate pools and cannot speak to judges.

Contributions.

Preprint.

- **The Atlas**: one factor model fitted to 173 SWE-bench Verified submissions, 126 MathArena configurations over 23 competitions and 165 LiveBench models, with intervals for every systematic share and held-out tests of what it predicts about sets of systems: coverage, the all-wrong tail and vote accuracy.
- **Depth on public data**: 19 pairs of one model at two adjacent reasoning settings, pooled with random effects, and a dollar-matched comparison of one deeper call against a wider vote (post hoc).
- **Preregistered API studies**: a Phase 1 case where one reasoning call wins, and an anchor study on 533 sealed items with two confirmatory tests (AH2: the first effort step against the nearest-dollar vote; AH3: the factor model's vote prediction), read once with frozen code.
- **Process**: sealed items, single reads, preregistered stop rules and 115 independent audits, for $196.64 of API spend in total. Every number in this paper is generated from a committed file and carries a status (Appendix **??**).

2 Definitions

We define each term in words first. **Width**: more calls of the same setting, or of different systems, combined by a rule (a plurality vote) or a selector. **Depth**: more reasoning per call (a higher effort or thinking budget) or a stronger tier. **Verification**: choosing among candidate answers, by a judge model, a self-check, execution, or a label-free rule such as Dawid–Skene Dawid and Skene [1979].

The factor model. System s answers item i correctly with probability $P_{si} = \sigma(a_s - \beta_s b_i)$, $\sigma(x) = 1/(1+e^{-x})$. The *ability* a_s is how good the system is overall; the *difficulty* b_i is how hard the item is for everyone, the shared part of failure (scaled to mean 0 and standard deviation 1); the *failure slope* β_s is how sharply the system's success falls with difficulty. Level L0 fixes $\beta_s = 1$. A *family shock* $u_{f(s),i}$ adds a shared draw for systems of one vendor; *call luck* is what remains, identified only from repeated calls. Fits are penalised logistic regressions (ridge 0.1), so items every system gets right or wrong keep finite difficulties.

Derived quantities. The *systematic share* $\mathrm{var}(b)/(\mathrm{var}(b) + \pi^2/3)$ is the share of latent variation in success that items share ($\pi^2/3$ is the variance of the logistic call-level noise). *Coverage* of a set of systems is the share of items at least one member gets right: the ceiling of any rule that returns one member's answer, as Chen [2026] also show; the *all-wrong rate* is one minus coverage. The *vote ceiling* of one system is the share of items whose most common answer is right, the limit of infinitely many repeats. *Selection efficiency* $\eta = (\text{selected} - \text{best single})/(\text{coverage} - \text{best single})$ is the share of the room above the best member that a selector reaches. A *consensus inversion* is a label-free rule rating a weaker system above the best one. *Calibration by dilution*: adding near-random votes shrinks a panel's calibration gap without improving its ranking (F13 in our findings ledger).

Statuses. CONFIRMATORY: a preregistered test in a Holm family, read once on sealed items. PREREGISTERED-DESCRIPTIVE: a preregistered quantity, read once, no pass/fail. EXPLORATORY: on sealed items, not preregistered as a test. POST HOC: chosen after the data were read, or on data not sealed for it; all public-data analyses are post hoc in this sense, as are numbers computed after a read. Post hoc results are many and uncorrected for multiplicity; we read them as descriptions, not tests.

3 The Atlas: what public data say about width and depth

Data. SWE-bench Verified Jimenez et al. [2024]: 173 submissions on 500 tasks, one run each, 39 with per-task cost and step counts (mini-SWE-agent). MathArena Balunović et al. [2025]: 126 configurations on 733 final-answer problems from 23 competitions, usually four calls each, with answers grouped into classes by the grader. LiveBench White et al. [2024]: 165 models on 228 items of three binary tasks, one call each, without answers or costs. Each benchmark is fitted on its own. All Atlas numbers are POST HOC.

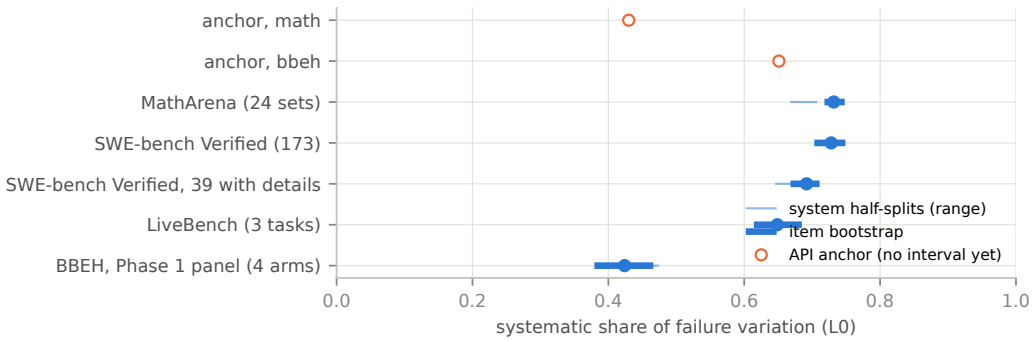


Figure 1: Systematic failure share per source (L0): item-bootstrap 95% interval (thick) and the range over system half-splits (thin; each half fitted on its own). The anchor's shares (hollow) await the gold files for an interval.

3.1 Failures are mostly shared

The systematic share is 0.73 on SWE-bench Verified (item bootstrap 0.70–0.75; over 40 random halves of the systems, 0.71–0.73), 0.73 on MathArena (0.72–0.75) and 0.65 on LiveBench (0.61–0.68); our Phase 1 BBEH panel of four arms sits lower at 0.42 (0.38–0.47; Figure 1). On the 39 SWE-bench submissions with per-task data, difficulty alone implies 98% of the mean pairwise failure correlation (0.502 of 0.512); two systems of one company add only 0.026 of residual correlation. All 39 fail 8.6% of tasks, against 0.0% if they failed independently.

3.2 Width is capped by the shared tail

Coverage and the all-wrong tail. Fitting difficulties on half the systems and each held-out system's ability on half the tasks, the model predicts the coverage of held-out 2–5 system sets on held-out SWE-bench tasks within 1.5–1.8 points; independence misses by 11–20. Chen [2026] report that a calibrated one-factor copula underprices the all-wrong tail by about 2.5× on open-ended mathematics. We give the factor model and the copula the same information, the calibration tasks only: the factor model draws an unknown test task's difficulty from the calibration tasks' difficulties, the copula fits its correlation to the calibration tasks' pairwise failures. For all 87 held-out SWE-bench systems on 250 held-out tasks the observed all-wrong rate is 7.6% (Clopper–Pearson 4.6–11.6); the factor model predicts 5.9% (observed over predicted 1.29, 0.68–1.84), the copula 3.5% (2.15×) and independence 0.00% (Figure 2). The factor model's free distribution of difficulties carries the tail that a Gaussian latent factor misses. On the small MathArena sets (15–26 test problems) both models underpredict the 5-system tail, on 2 and 2 of 6 sets; the math tail is unresolved.

Votes of one system. A vote of two calls cannot gain in expectation (a right–wrong split is a tie), so gains start at three. The factor model's *vote rule* predicts a k-vote from one-call success p, the share z of failures that abstain (abstentions are non-votes) and the chance q that two wrong answers coincide. Cross-fitted on 695 (system, set) cells with three calls in 24 sets, its mean absolute vote-gain error is 2.0 points against 2.2 for the zero-gain predictor "votes add nothing"; predicted and observed gains correlate at 0.46, and the gain's squared error is smaller by 2.4 points^2 (0.6 to 3.4; item noise adds the same variance to both and cancels). But the rule overstates gains (4.7 predicted, 2.9 observed where it predicts 3 or more), and on the 3-vote accuracy itself it is worse than zero-gain (mean absolute error 4.4 against 4.0 points; squared error larger by -4.2 points^2, -9.1 to -1.6). The rule ranks where votes pay; it does not yet price them.

Votes across systems. Votes pay most when weak systems with different strengths are mixed. On 1,577 panels of three and 1,585 of five MathArena systems with one-call accuracy 0.2–0.7, the plurality of one call each gains 5.7 and 12.5 points over a random member. Cross-fitted, the factor model predicts the panel vote with bias 0.4 and 0.4 points and absolute error 6.4 and 6.6, against 8.2 and 13.6 for zero-gain and 11.9 and 16.2 for independence; it is closer than zero-gain in 4 and 4

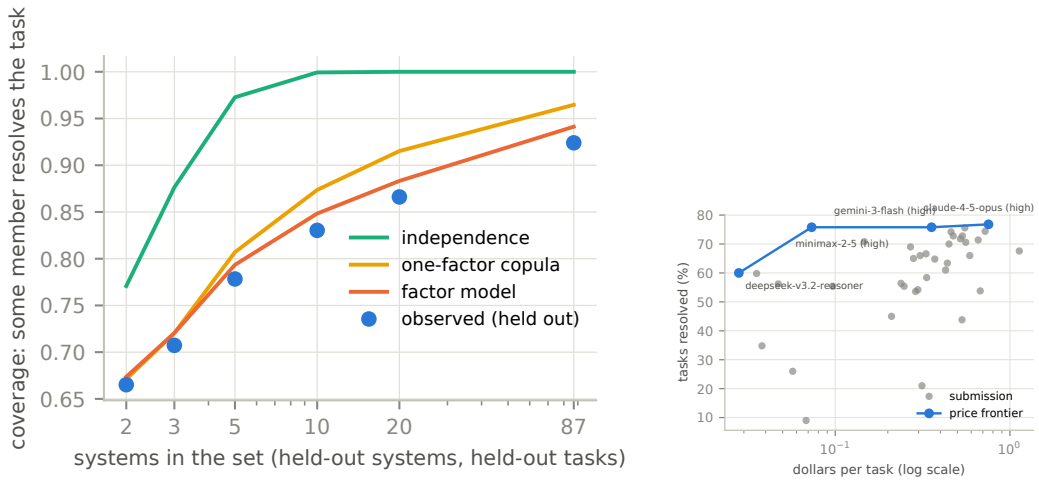


Figure 2: Left: coverage of sets of held-out SWE-bench Verified systems on held-out tasks, observed and predicted from the calibration tasks only (P8). Right: tasks resolved against dollars per task, SWE-bench Verified submissions with costs; the price frontier in blue (A7).

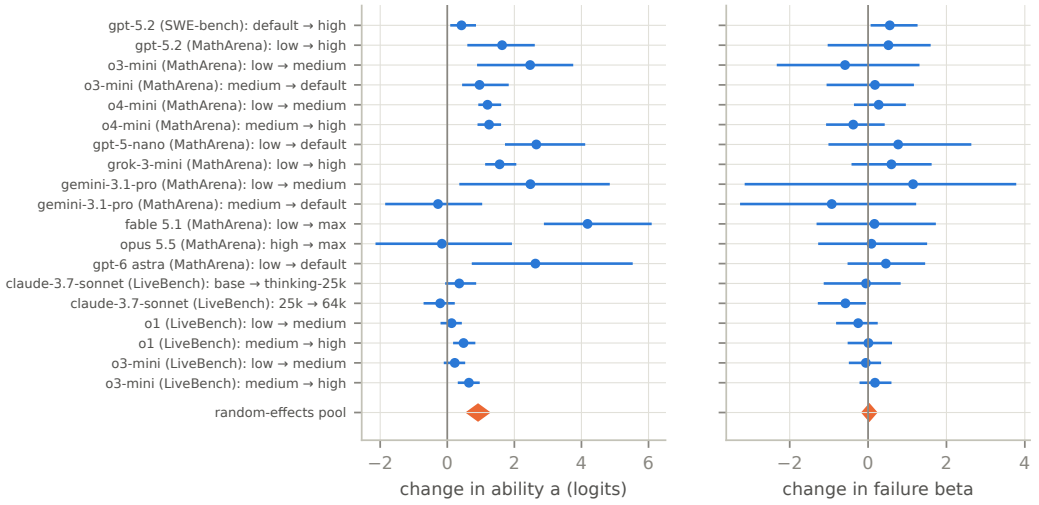


Figure 3: One model at two adjacent reasoning settings (P2): change in ability a and in failure slope β with item-bootstrap 95% intervals, and the DerSimonian–Laird random-effects pool (diamond).

of 5 sets, and closer than independence in 4 and 5. Where composition matters, the model prices it without bias; independence overstates it and zero-gain ignores it.

3.3 Depth moves ability, not the failure pattern

For each of 19 public pairs of one model at two adjacent reasoning settings (11 models, some on two benchmarks; difficulties from every other system), we fit both settings' ability and slope on the items both answered (Figure 3). Pooled with random effects, ability rises by 0.92 logits (0.59–1.24; between-pair variance 0.35); the slope changes by 0.03 (-0.14–0.21; between-pair variance 0.00). One pair per model and benchmark (13 pairs) gives 1.21 and 0.11. Held out, the higher setting's calls are predicted better by the lower setting plus one shift than by a refit that may also change the slope (log loss per call 0.0041 in favour of the shift, 0.0022–0.0061) and far better than by its base rate (0.192 nats per call). The difficulties are the population's, so this says the higher setting fails the items that are hard for everyone; the lower setting's own item outcomes plus a shift predict no better

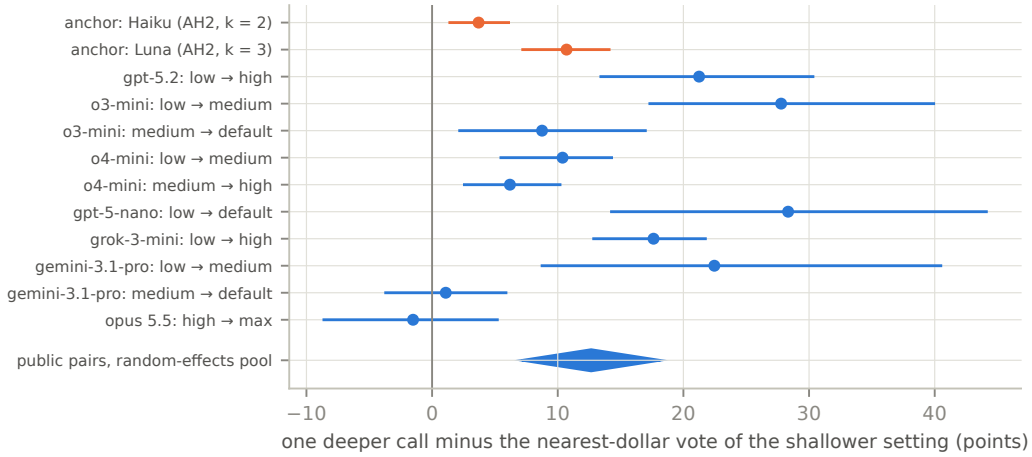


Figure 4: One deeper call minus the nearest-dollar vote of the shallower setting: the anchor's confirmatory AH2 (orange, fresh sealed items, the first effort step of two cheap models) and public MathArena pairs (blue, post hoc) with item-bootstrap 95% intervals and their random-effects pool.

(0.007, -0.017 to 0.030). Two pairs left out after a look at output tokens (one setting of unknown order, one different checkpoint) are shown in the extension report and were never pooled.

Depth against width at equal dollars (post hoc). For 10 MathArena pairs whose lower setting has three or more calls, one call at the higher setting minus the vote of the lower setting with the nearest dollars per item (the 3-vote in every pair) is 12.7 points (7.1–18.2); on the 7 pairs priced within a factor of 1.5 it is 11.9 (6.0–17.7; Figure 4). The step pays where a model starts low and about nothing at the top of its range: +1.1 points for Gemini 3.1 Pro medium to default and -1.5 for Opus 5.5 high to max. We logged, before it is read, that this predicts about zero for the preregistered CD2 test of a top Haiku effort step. Price matters as much: on SWE-bench Verified MiniMax M2.5 (high) resolves 75.8% at $0.073 per task against 76.8% at $0.754 for Claude Opus 4.5 (high), 10× the price for one point (Figure 2, right).

4 Preregistered API studies

Phase 1: the saturated case. On 300 sealed BBEH items Kazemi et al. [2025], one GPT-6.1 Sol call at low reasoning was right 97.3% of the time at $0.0062 per call; Claude Sonnet 5.5 without reasoning 72.0% at $0.0139 (PREREGISTERED-DESCRIPTIVE). Width bought little: 5 Haiku calls are worth 1.84 independent calls (repeat correlation 0.43), their plurality reached 48.3%, a 12-call mixed panel 58.0%, and none of 276 cascade settings came within 2 points of Sol at 0.8× its dollars (EXPLORATORY/POST HOC). The confirmatory cascade test (H7) passed non-inferiority to Sonnet (-0.7 points) only by escalating 78% of items. Sol's accuracy is not answer memory: 40 of 40 perturbed items right. The Sol–Sonnet gap mixed vendor and reasoning, which motivated the anchor.

The anchor. 440 BBEH items (linguistic puzzles, cartoon captions, word sorting, causal judgement) and 93 arXivMath problems were sealed before any call. Arms: Claude Haiku 5.5 (off, low, high), Sonnet 5.5 (off, low), GPT-6 Luna (none, low, medium), GPT-6.1 Sol (low, medium), most with three calls per item; dollars are measured at list prices. Each block was read once with frozen code; Holm at 5% over the two tests that kept power of at least 0.70 (AH2, AH3).

*AH2 (*CONFIRMATORY*, passes;* $p = 0.001$*).* Raising a cheap model's reasoning one step beat repeating its no-reasoning call at the nearest measured dollars: Haiku at effort low against the Haiku-off 2-vote by 3.7 points (1.3–6.2), Luna at reasoning low against the Luna-none 3-vote by 10.7 (7.1–14.2); the deeper call cost 1.17× and 1.38× as much. For Haiku the 2-vote equals one call in expectation, so this shows that effort low beats effort off at slightly more dollars, not that depth beats a wider vote; against the dearer 3-vote the difference is 1.8 (-0.7–4.3), not tested. The effect sits mainly in word sorting (and linguistic puzzles for Luna; computed after the read). The decision as frozen: for cheap

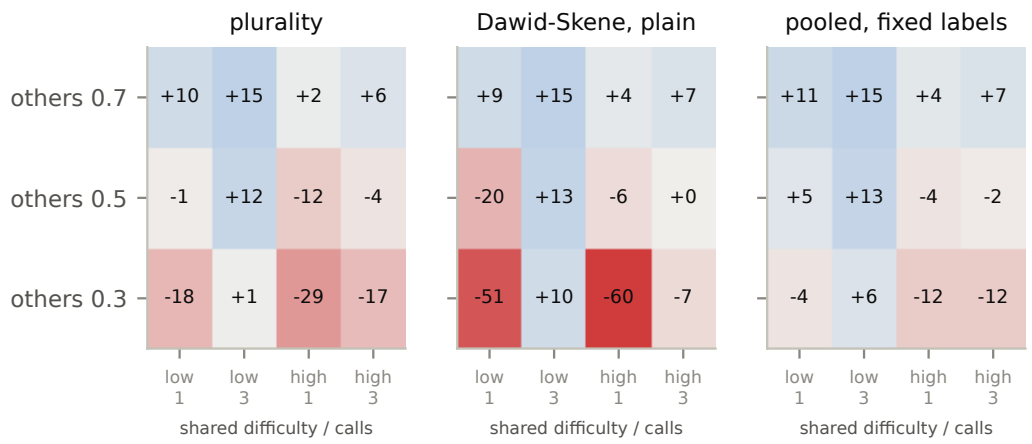


Figure 5: Label-free selection, simulation map (P4): gain over the best system (points) when one system at 0.85 sits among others at 0.3, 0.5 or 0.7, at low or high shared difficulty (systematic share 0.07 or 0.55), with one or three calls per system; averaged over 3, 5 and 9 systems.

models on these items, buy the first effort step before width. It covers neither higher steps nor other tasks.

*AH3, the vote rule (*CONFIRMATORY*, passes).* Cross-fitted on single calls, the factor model predicted the six 3-call arms' 3-vote accuracy with mean absolute error 0.8 points (upper bound 2.4) and their gain over one call within 0.7 (upper bound 1.3). The gains were small (0.4 to 2.6 points) and zero-gain also meets the limits (gain error 1.5 against 0.8), so AH3 shows accuracy where votes gain little; P1 above finds the same rule overstating larger gains.

*AH1 and AH4 (*EXPLORATORY*).* Sonnet's effort-low setting closed none of the 11.1-point gap to Sol low ($R = 0.02$, -0.32 to 0.26), so the vendor–reasoning confound stays open. The Sol judge's selection efficiency was 0.85 on easy items and 0.14 on hard ones (difference 0.71, 0.41–1.05), the predicted direction, with a ratio whose hard-half denominator is small.

The depth study. A preregistered study of Haiku at every reasoning effort (research/v4/cd/read.md) finds that accuracy gains flatten by the second-highest effort; the drop at the highest effort is within its interval and coincides with calls cut at the output limit.

5 Verification

Label-free selection. With repeats, Dawid–Skene rated Sol at 62.5% on Phase 1 against a true 97.3% (EXPLORATORY); on the anchor its pooled variant put Sonnet at 79% against a true 45%, and its answers (47.5%) stayed below one Sol low call (56.2%). A simulation map (Figure 5) shows when this happens: with one strong system among weak ones and one call each, plain Dawid–Skene ranks a weak system first in 90% of panels; a fixed label space removes the inversion in the simulation (0%) but still loses 8.0 points to the strong system. On MathArena the fixed label space does not help: beside a lone strong system it ranks another system first in 62–71% of panels, against 42–63% for plain Dawid–Skene. At matched calls (three systems with one call each, against the best system's own 3-vote, the best chosen with labels) every policy loses beside a lone strong system (plurality -34.0 points, plain Dawid–Skene -25.7) and none gains among competent ones (plurality -1.8, plain Dawid–Skene -1.0, fixed labels -0.3).

The anchor's judge, and why it cannot test judges. On the anchor's BBEH items the Sol low judge, choosing among the answers of two cheap pools, reached 52.3%: 13.3 points above the best pool arm (9.8–15.8), selection efficiency 0.70 (0.57–0.80). Sol low answering alone scored 60.6%, so the judge sat 8.3 points below it (-12.0 to -4.8) at 1.6× the dollars (Figure 6; POST HOC, computed after the read). But the pools covered only 58.0% of the items: even a perfect selector over them would have lost to Sol alone. The comparison measures the pools, not the judge's signal; a test of

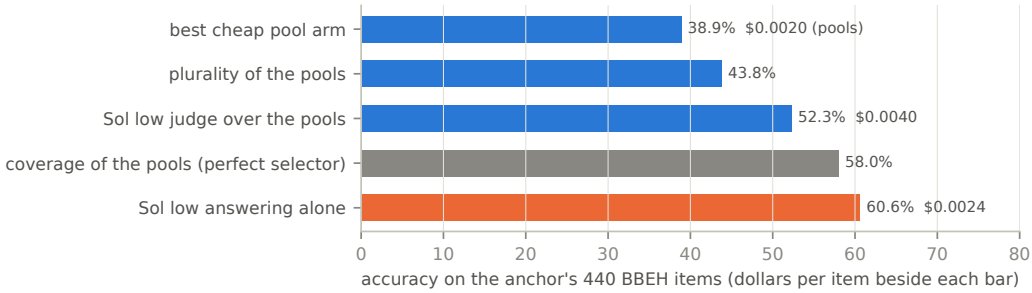


Figure 6: The anchor's BBEH verification read: a judge over cheap pools against the judge model answering alone. The pools' coverage (grey) is below the judge model's own accuracy, which caps every selector over them.

judges needs pools whose coverage exceeds the judge model's own accuracy. On math the judge was capped by 22.6% coverage.

The price of verification on public data. A perfect selector over random 2–5 system SWE-bench sets adds 4.8–8.3 points over the best member, at \$0.062–\$0.163 of extra spend per point per task. With the anchor's best measured efficiency, a judge over such a set beats the most accurate single submission at no more of its dollars in at most 10% of sets (a perfect selector in at most 13%, judge calls unpriced). On MathArena the room is 1.0–4.5 points because the top systems are close to perfect.

Self-checks and execution. Sol accepted 97% of its own anchor answers (AUROC 0.69); Haiku 66% (AUROC 0.78). On SWE-bench the step count of a run ranks its own successes with median AUROC 0.68; a cascade that escalates on step count never escalates once its threshold is fitted. Execution-based checks on code are part of the open-weight testbed.

6 The open-weight testbed (design; results pending)

The testbed varies precision (BF16, W8, W4, W3), size (Qwen3.5 2B, 4B, 9B) and thinking budget (off, 1k, 4k) with several samples per item on 363 items (200 BBEH, 63 competition math, 100 LiveCodeBench), so width, depth and verification can be bought at measured GPU cost. Its preregistration (T1–T5) has power by simulation of 0.94 (T1: the W4 loss survives voting on hard items), 1.00 (T2), 1.00 (T3: depth against cost-matched width), 1.00 (T4: judge coupling to difficulty) and 0.65 (T5, exploratory: compositional prediction) (DESIGN). The pilot is estimated at \$4.1 of GPU time. No GPU has run yet; no result is reported.

7 Related work

Shared failure and its ceiling. Chen [2026] show that any routing, voting or cascade policy returning one model's answer is bounded by one minus the share of queries every model fails, certify that share from a graded set, and find that a one-factor copula underprices it on open-ended mathematics; they exclude thinking variants and do not hold dollars fixed. We take the ceiling as given and ask how depth moves it and at what price. Kim et al. [2025] measure correlated errors across hundreds of models without an item-difficulty term; Turkmen et al. [2026] prove an error floor under a Gaussian latent factor; Denisov-Blanch et al. [2026] find that polling does not improve truthfulness because errors are shared. Ge et al. [2026] fit an ability-minus-difficulty model to agentic coding leaderboards to predict tasks and agents; we use the same scale for composition.

Width. Chen et al. [2024] show that vote accuracy can rise and then fall with more calls of one model on a mix of easy and hard queries; they count calls, not dollars. Chen et al. [2025] assign different models to the modules of a compound system with a labelled search, which can beat any single model; this is role allocation, not parallel width.

Verification and cascades. Singhi et al. [2025] find self-consistency more compute-efficient than generative verification at most practical budgets, with verification ahead only at many times the compute; our comparator is the judge model's own answer at measured dollars. Conformal cascades Dou et al. [2026] need labelled calibration data; C3PO Valkanas et al. [2025] anchors a label-free cascade on the strongest model's answers and so inherits its errors. Learn-then-Test Angelopoulos et al. [2021] gives the risk control we plan for a runtime router.

8 Limitations

Models and prices are those of October 2026, and public leaderboards mix scaffolds and dates. Most Atlas results are post hoc and uncorrected for the many comparisons made. Contamination cannot be ruled out: BBEH has been public since 2025, so "fresh" means new to this project, and LiveCodeBench predates the testbed's models. The API studies are small (533 items in the anchor) and use two vendors; the AH2 decision covers the first effort step for cheap models on these items only, and depth beyond it awaits CD2. The anchor's judge test was capped by its pools. Answer format moves the difficulty axis Chen [2026]; our anchor tasks are free-form. The all-wrong tail on small math sets is not resolved. The anchor's systematic shares have no interval until the gold files are restored.

9 Conclusion and next steps

On the data we have, failures are mostly shared, and a factor model with one free difficulty per item prices the shared tail on code better than a one-factor copula given the same data. Reasoning moves a system along the shared difficulty axis without changing which items it fails. The first reasoning step of cheap models beat a dollar-matched vote of their no-reasoning calls on sealed items; whether depth beats width beyond that step is open, and our public pairs predict about zero at the top of a model's range. Selection by agreement did not beat the best system's own vote at matched calls, and a test of judges still needs pools that cover more than the judge answers alone. The next steps (roadmap in research/v4/roadmap.md): 2,000+ items across 8 or more task families including agentic ones, with a time-split replication; a third vendor used once on the frozen design; the open-weight testbed; batch and cache pricing; and a budget allocator that turns a 30–50 item pilot and a price table into a depth, width and verification mix with predicted accuracy, cost and uncertainty.

References

Anastasios N. Angelopoulos, Stephen Bates, Emmanuel J. Candès, Michael I. Jordan, and Lihua Lei. Learn then test: Calibrating predictive algorithms to achieve risk control. *arXiv preprint arXiv:2110.01052*, 2021.

Mislav Balunović, Jasper Dekoninck, Ivo Petrov, Nikola Jovanović, and Martin Vechev. MathArena: Evaluating LLMs on uncontaminated math competitions. *arXiv preprint arXiv:2505.23281*, 2025.

Josef Chen. When does combining language models help? a co-failure ceiling on routing, voting, and mixture-of-agents across 67 frontier models. *arXiv preprint arXiv:2606.27288*, 2026.

Lingjiao Chen, Jared Quincy Davis, Boris Hanin, Peter Bailis, Ion Stoica, Matei Zaharia, and James Zou. Are more LLM calls all you need? towards scaling laws of compound inference systems. *arXiv preprint arXiv:2403.02419*, 2024.

Lingjiao Chen, Jared Quincy Davis, Boris Hanin, Peter Bailis, Matei Zaharia, James Zou, and Ion Stoica. Optimizing model selection for compound AI systems. *arXiv preprint arXiv:2502.14815*, 2025.

A. P. Dawid and A. M. Skene. Maximum likelihood estimation of observer error-rates using the EM algorithm. *Journal of the Royal Statistical Society, Series C*, 28(1):20–28, 1979.

Yegor Denisov-Blanch, Joshua Kazdan, Jessica Chudnovsky, Rylan Schaeffer, Sheng Guan, Soji Adeshina, and Sanmi Koyejo. Consensus is not verification: Why crowd wisdom strategies fail for LLM truthfulness. *arXiv preprint arXiv:2603.06612*, 2026.

Yifan Dou, Shikan Lian, and Shibo Li. Conformal cascade: Distribution-free accuracy guarantees for multi-tier LLM inference. *arXiv preprint arXiv:2607.25018*, 2026.

Chris Ge, Daria Kryvosheieva, Daniel Fried, Uzay Girit, and Kaivalya Hariharan. Agent psychometrics: Task-level performance prediction in agentic coding benchmarks. *arXiv preprint arXiv:2604.00594*, 2026.

Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. SWE-bench: Can language models resolve real-world GitHub issues? In *International Conference on Learning Representations (ICLR)*, 2024.

Mehran Kazemi et al. BIG-Bench Extra Hard. *arXiv preprint arXiv:2502.19187*, 2025.

Elliot Kim, Avi Garg, Kenny Peng, and Nikhil Garg. Correlated errors in large language models. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2506.07962.

Nishad Singhi, Hritik Bansal, Arian Hosseini, Aditya Grover, Kai-Wei Chang, Marcus Rohrbach, and Anna Rohrbach. When to solve, when to verify: Compute-optimal problem solving and generative verification for LLM reasoning. In *Conference on Language Modeling (COLM)*, 2025. arXiv:2504.01005.

Yigit Turkmen, Baturalp Buyukates, and Melih Bastopcu. Don't always pick the highest-performing model: An information theoretic view of LLM ensemble selection. *arXiv preprint arXiv:2602.08003*, 2026.

Antonios Valkanas, Soumyasundar Pal, Pavel Rumiantsev, Yingxue Zhang, and Mark Coates. C3PO: Optimized large language model cascades with probabilistic cost constraints for reasoning. *arXiv preprint arXiv:2511.07396*, 2025.

Colin White et al. LiveBench: A challenging, contamination-free LLM benchmark. *arXiv preprint arXiv:2406.19314*, 2024.

A Reproducibility

Preregistration. Every paid run carried a committed `prereg:` block (question, prediction, kill rule, decision) that code checked before the run could spend; the full texts are in the repository's research/ folder. Tests whose simulated power at the chosen n fell below 0.70 were reported as exploratory.

Audits. An independent audit read every design and every paid run's outcome before the next step: 115 verdicts (31 GO, 83 ADJUST, 1 STOP); every ADJUST was resolved before further spend.

Cost. Total API spend $196.64 over 502 runs, under hard per-account caps enforced in code. Every call is stored, so every team in this paper is rescored by replay at no cost. The per-run ledger and the status of every number in the paper are generated in the repository (paper/runs_table.tex, paper/numbers_table.tex).